A Multi-Hazard Assessment of 2,696 US Data Centers

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Background

- AI and cloud demand has concentrated US computing into a few dozen metro clusters that face earthquakes, wildfire and flooding.
- Existing sources are proprietary, or give city-level coordinates never checked against a building.
- We built a free, public, building-level inventory of the contiguous US, and tested how much the answer moves when a coordinate is wrong.
- We measured exposure, meaning whether a site lies in a hazard zone. We did not model damage, so nothing here is a loss estimate.

Limitations

- Coverage is uneven. 83% of our Virginia sites come from OpenStreetMap against 48% of our California ones, so part of the gap between states reflects who did the mapping.
- Hail and wind come from eyewitness reports, so within a state their rates track how built-up the surroundings are (rank correlation 0.39 and 0.34). We left both out, along with lightning, tornado and hurricane.
- No public source lists power capacity, so a single server room counts the same as a 100 MW campus.
- 40 sites shown as clear sit outside FEMA's mapped extent and 8 outside the wildfire raster, so 35% is a floor.

Materials and Methods

We merged OpenStreetMap, PeeringDB and Wikidata into one record per site across the contiguous US, matching on name and distance, with written evidence kept for all 194 merges.

Against FEMA USA Structures footprints, 1,681 coordinates fall inside a building, 993 match the nearest and 22 have no match. A second non-OSM source confirmed 1,546, and we re-checked the 1,150 least certain with an AI-assisted pipeline we then reviewed.

Our first seismic layer failed validation against USGS reference values (22 of 150 sites within 10%), so we replaced it:

Earthquake	USGS ASCE 7-22 shaking, 2% chance in 50 yr
Wildfire	USFS Hazard Potential 270 m, max in 2.4 km
Flood	FEMA NFHL layer 28, SFHA flag
Water stress	WRI Aqueduct 4.0 (a supply limit, reported apart)
Uncertainty	500 relocations per site, 10-500 m by precision

A site counts as exposed if it lies within 2.4 km of land rated High or Very High for wildfire, inside a FEMA Special Flood Hazard Area, or at peak ground acceleration of 0.30 g or above. The 2.4 km radius is the 1.5 mile ember-cast distance.

Results

2,696	954	952	563
facilities located	face a mapped hazard	in a high water-stress basin	water-stressed only

The hazard map and the water map barely overlap. Counting water stress takes the fleet from 35% to 56% constrained.

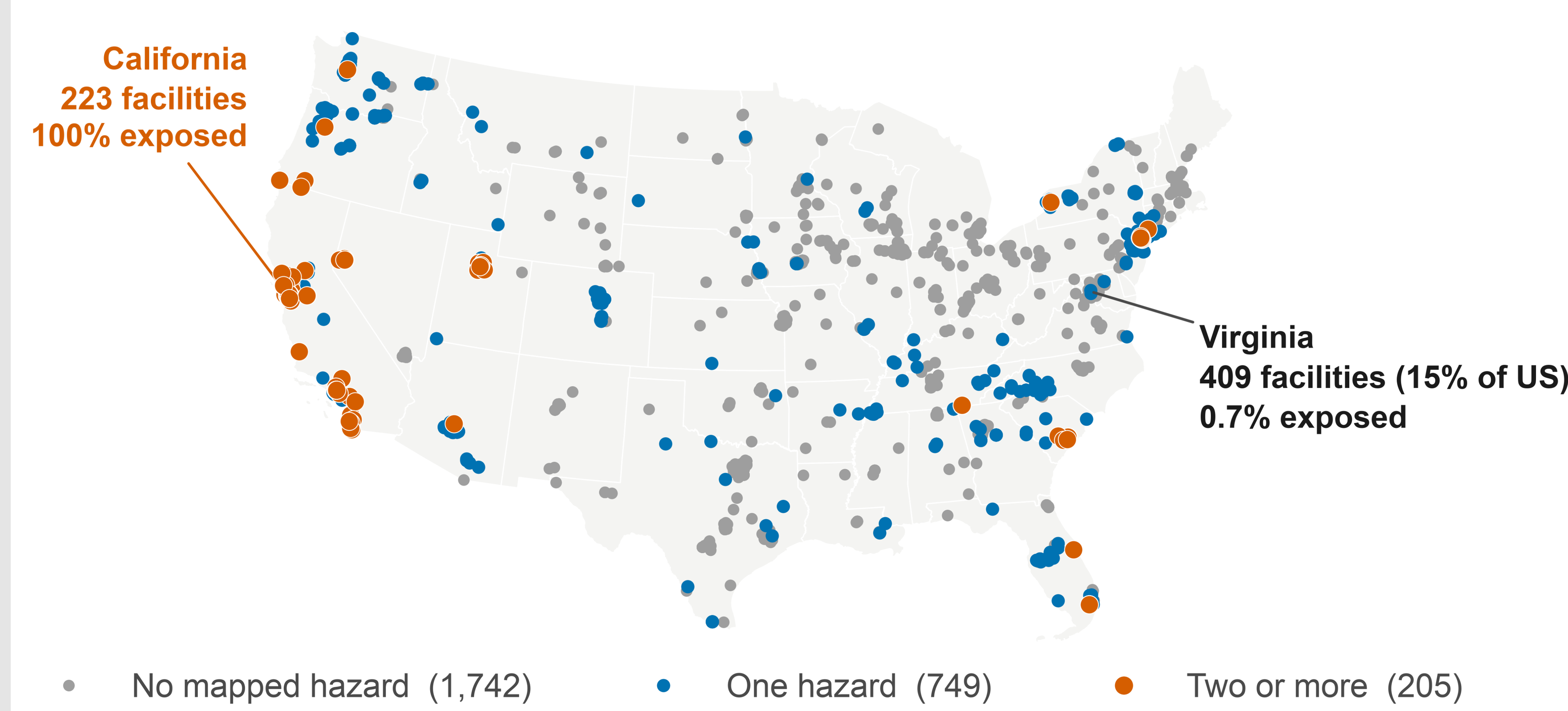

Fig 1. One third of US data centers face a mapped hazard. Of the 205 facing two or more (orange), California holds 119, with smaller clusters in Utah, Nevada and New Jersey.

952 sites lie in a high or extremely-high water-stress basin, almost exactly the 954 facing a mapped hazard, but only 389 are the same sites. Water stress lands hardest where the hazard map says clear: Virginia is 0.7% hazard-exposed and 31% water-stressed, Illinois 2.7% and 95%.

Which hazard dominates depends on the region. New Jersey reaches 100% through wildfire alone and 0% through earthquake, California 96% earthquake. Nationally 646 sites are exposed to wildfire, 448 to earthquake and 71 to flood, and 205 face more than one.

Widening the wildfire buffer to 5 km takes that count from 646 to 953, a 48% jump.

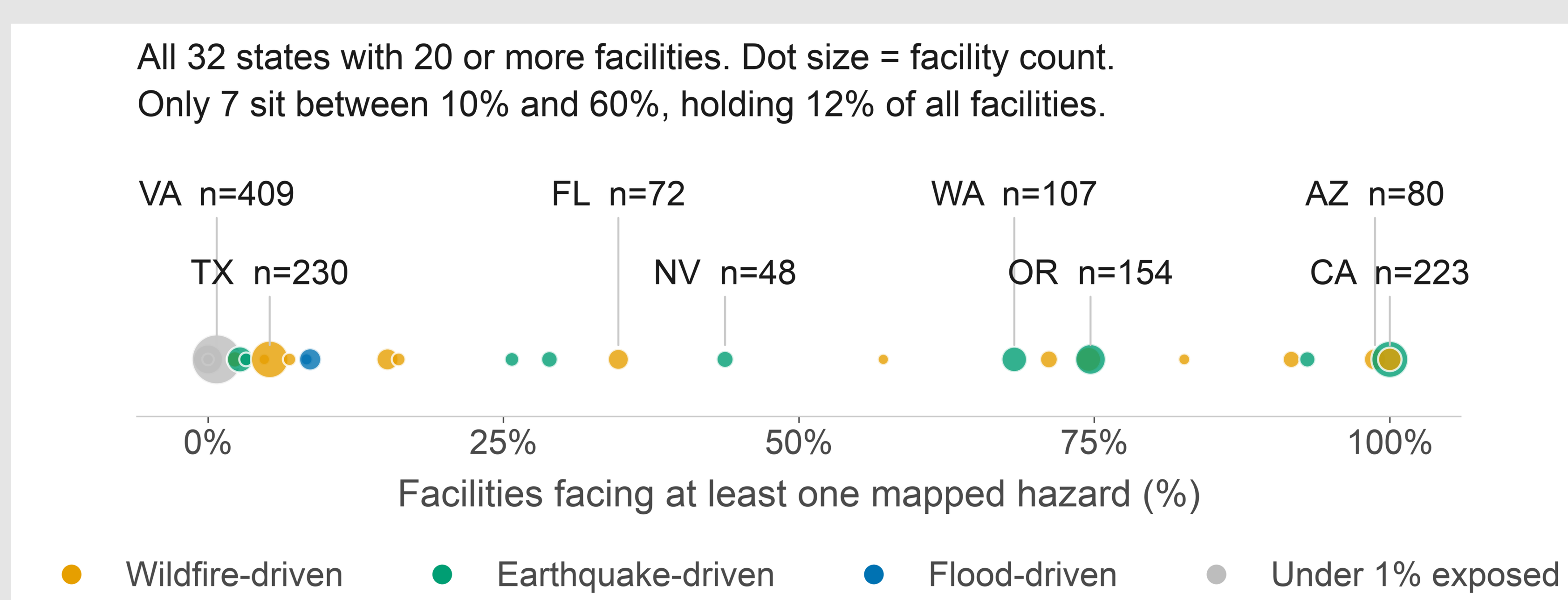

Fig 2. All 32 states with 20 or more sites, coloured by dominant hazard.

Conclusions

- The 35% headline is the least useful number here. Only 7 of 32 states with 20 or more sites fall between 10% and 60% exposed.
- Water stress is not a natural hazard, but it constrains the same buildings, and it is high where the hazard map says clear.
- New Jersey and California are both 100% exposed for opposite reasons, so one national score would hide what a site faces.
- Next: fragility curves, to turn exposure into loss.

Two checks on our choices

Sampling across a building's whole footprint instead of its centre changed the answer for 29 of the 326 sites where both could be compared. Taller buildings looked more exposed nationally, but that vanished within states, so it was geography and not height.

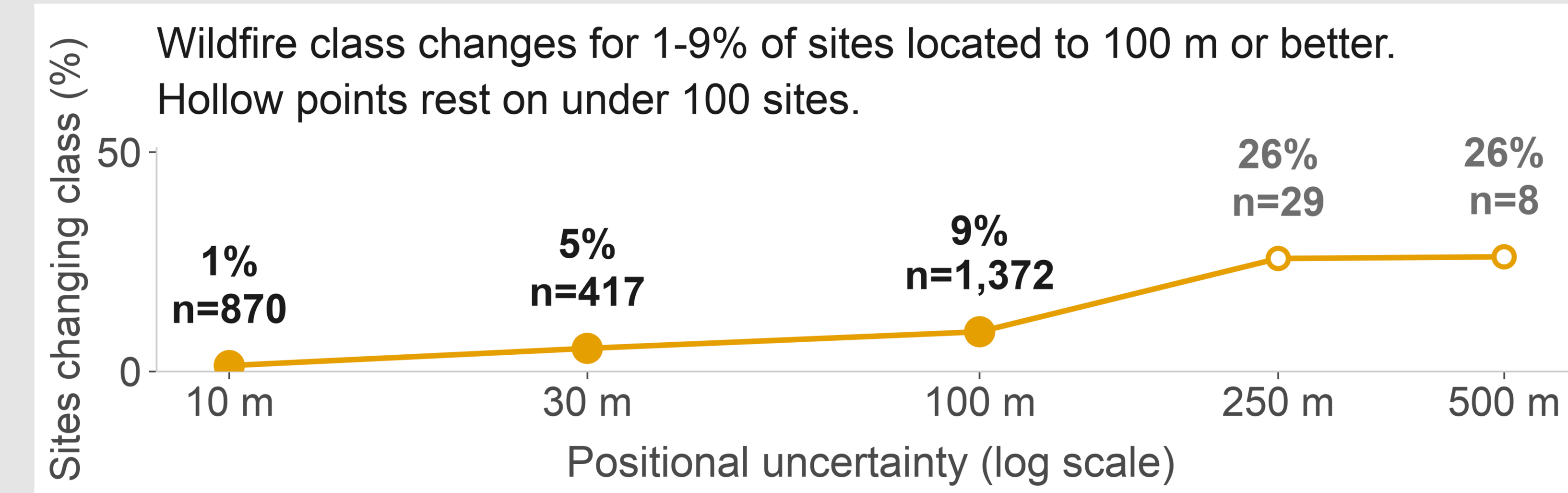

Fig 3. Moving each coordinate 500 times flips the wildfire class for at most 9% of sites placed within 100 m.

Major Citations

Dillon, G.K. (2023) Wildfire Hazard Potential for the United States, 270-m, v2023. USDA Forest Service. doi:10.2737/RDS-2015-0047-4

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FEMA and ORNL (2024) USA Structures. gis.fema.gov

WRI (2023) Aqueduct 4.0 Water Risk Atlas.

Acknowledgements

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